

# Volume 0 — Exact Witnesses at the Physics Boundary

The former T4–T6 "Projection Craft" papers, rebuilt as standalone R Theory.

**What this volume is.** Three books that take the canonical half-angle calculus of [Book 0](#) to the boundary of accepted physics — relativistic kinematics, the Dirac spinor, weak helicity, Coulomb structure, color, bound states, and canonical spin–geometry — and report exactly what is found there: exact coordinate correspondences, exact conditional theorems, honest obstructions, and no new physical law.

**Why it is standalone.** The source material was written as "Projection Craft" essay-books T4, T5, T6, framed as continuations of the T-paper series. That framing is not carried over. Each book here states its own dependencies, its own imports, and its own status ledger from scratch. Nothing is cited to a T-paper authority; every result is presented with the mathematics or the import that supports it.

**Dependencies.** All three books inherit the Book 0 core: the reciprocal primitive calculus, the one-generator reduction, the Clifford/bilinear structure, and the coframe geometry. Books 21 and 22 additionally depend on the internal paper *Reciprocal Half-Angle Coordinates in the Radial Dirac–Coulomb Problem* (read and algebra-checked 2026-09-19), which supplies the conditional spinor-ratio identities both books use. Named physics imports — special relativity, the Dirac equation, V–A, QCD, Coulomb/Dirac–Coulomb theory, Maxwell, Einstein–Hilbert, Einstein–Cartan — are declared per section, never smuggled.

**Provenance note (2026-09-19).** The T6 source preamble credits the T5 source with a "Yang–Mills source-free-vacuum obstruction" result — a static homogeneous constant noncommuting ansatz whose curvature the source-free Yang–Mills equations force to vanish. A full-text search of the T5 source (v2 reconstruction) finds no such result: the only occurrence of "Yang–Mills" is the QCD import line. The attribution describes the original T5 research draft, not the reconstructed text, and it is **not repeated** anywhere in this volume. If the original computation is located and verified, it will be added with its source cited; until then the claim does not exist in R Theory.

## The shared discipline

Every book in this volume closes with an operational prediction-difference audit. For a fully specified Projection-augmented model  $MP$  against its accepted reference  $M0$ , on a common measurement contract,

$$\Delta_{op}(MP, M0) = \{ c \in C_{exp} : P_P(\cdot | c) \neq P_\theta(\cdot | c) \}$$

is the set of experimental configurations where the predictions differ. Three terminal statuses are distinguished: **Class A** — fully specified and  $\Delta_{op} = \emptyset$  (operationally equivalent); **Class B** — incomplete,  $\Delta_{op}$  undefined (an open ansatz is not a nonempty prediction set); **Class C** — fully specified with  $\Delta_{op} \neq \emptyset$ . An invertible reparameterization can never leave Class A; that is proved, not assumed.

The null hypothesis retained by all three books is **C0**: *Projection Craft changes representation but not physics*. It is falsifiable — a future fully specified Class-C model would defeat it — but undefeated by everything in this volume. Exactness without promotion is the theme: the correspondences are real mathematics, and they are not new laws of nature.

**PDF snapshot.** Every issue in this volume is closed and every claim is proven under the status discipline above, so a frozen PDF is offered: [Volume 0 complete \(PDF\)](#) — overview plus the three books in reading order. Individual pages: [overview](#), [Relativistic and Spinor Witnesses](#), [Fermion Composition, Color Closure, and Bound-State Geometry](#), [Canonical Spin–Geometry and the Primitive Symplectic Atlas](#). Snapshot generated 2026-09-19; content identical to these pages.

## The books

### Relativistic and Spinor Witnesses

The kinematic witness ( $\sin \chi = \beta$ , with  $sxp/cxp/crx$  as the Dirac half-angle ratio and rapidity weights), the two-state spinor completion and its independent phase, the V–A helicity weights after explicit chiral import, the beta-decay/Coulomb two-angle bridge, and the adversarial electromagnetic comparison — Jones, duality,

Newman–Penrose, Veronese, conformal — ending in the full  $\Delta_{\text{op}}$  audit. Every fully specified construction:

Class A.  $\Delta_{\text{op}} = \emptyset$ .

[Read Book 20](#) →

## Fermion Composition, Color Closure, and Bound-State Geometry

The composition firewall, the three-state carrier rebuilt in the correct logical order ( $C^3 \rightarrow A_2 \rightarrow \text{su}(3)$ , tetrahedral metric demoted to compatible realization), the QCD import with its spatial firewall, the exact two-fermion invariant, the below-threshold coordinate and its conditional Dirac–Coulomb convergence, observational discipline with verified hydrogen/deuteron/muonium numbers, the frozen failure record (81/32, fifth lattice, musical commas), and a firewalled E8 appendix.  $\Delta_{\text{op}} = \emptyset$ .

[Read Book 21](#) →

## Canonical Spin–Geometry and the Primitive Symplectic Atlas

The volume's strongest exact result: after the declared radial two-component chart and the imported radial Dirac dynamics, the four primitives are canonical charts on one radial-spinor phase plane —  $p_{\text{sxp}} = 2F^2$ ,  $p_{\text{srx}} = -2G^2$ ,  $p_{\text{cxp}} = (F-G)^2$ ,  $p_{\text{crx}} = -(F+G)^2$ , common form  $\omega_{\text{spin}} = 4 \, dF \wedge dG$ . Around that core: the Prüfer canonical pair, the first-order spherical defect action, charge and angular momentum as canonical momenta, Kerr/Kerr–Newman half-angle identities with their guards, and the full does-not-establish ledger.

$\Delta_{\text{op}} = \emptyset$ .

[Read Book 22](#) →